

A Contractive but Not Completely Positive Map on $\mathcal{OS}(J_3)$

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2026-07-22

Status. Candidate counterexample, likely valid. The construction is existential but finite-dimensional: the separating operator can be chosen in $\mathbb{M}_3$.

Source and Conjecture

The source is Pankaj Dey, Atanu Dhang, and Mithun Mukherjee, *Matricial ranges, dilations, and unital contractive maps*, arXiv:2509.18027 [1]. Write

$$J_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathcal{OS}(J_3) = \text{span}\{I_3, J_3, J_3^*\}.$$

Conjecture 6.10, on page 21, states that for a unital self-adjoint preserving map $\varphi : \mathcal{OS}(J_3) \rightarrow \mathcal{OS}(X)$ with $\varphi(J_3) = X$, the following are equivalent:

1. φ is completely positive;
2. φ is contractive;
3. $\|\alpha X + \beta X^*\| \leq \sqrt{|\alpha|^2 + |\beta|^2}$ for all $\alpha, \beta \in \mathbb{C}$.

The conjecture is false. In fact, condition (iii) need not imply even 3-positivity.

Proof Intuition

The source paper already contains the obstruction, but it must be viewed through matrix-convex duality. Its Example 4.11 gives a matrix B_0 for which $B_0 B_0^* + B_0^* B_0 \leq I$, yet B_0 cannot be compressed from any matrix satisfying equality. The equality matrices

$$G_{a,b} = \begin{bmatrix} 0 & a \\ b & 0 \end{bmatrix}, \quad |a|^2 + |b|^2 = 1,$$

are exactly the elementary boundary tests used by the scalar norm condition.

Take the closed matrix-convex hull of these $G_{a,b}$. The matrix B_0 does not belong to it, since membership would itself give an equality dilation. The Effros–Winkler matrix Hahn–Banach theorem therefore produces a monic pencil which is positive on every $G_{a,b}$ but not at B_0 . Writing the two self-adjoint coefficients of this pencil as one operator X , positivity on all $G_{a,b}$ becomes exactly $\|\alpha X + \beta X^*\| \leq \sqrt{|\alpha|^2 + |\beta|^2}$. Failure at B_0 , however, says that the corresponding map on $\mathcal{OS}(J_3)$ is not 3-positive.

Indeed, the first inclusion in Corollary 6.8 is strict since $J_3 \in W_n(J_3)$ but $J_3 J_3^* + J_3^* J_3 \not\leq I_3$. However, it remains unclear whether the second inclusion in Corollary 6.8 is also strict. At present, we do not have an example demonstrating this. In fact, all our MATLAB computations confirm that the second inclusion is an equality. Therefore, we conclude this section with the following conjecture for further study in the future.

Conjecture 6.10. *Let $X \in \mathcal{B}(\mathcal{H})$. Suppose $\varphi : \mathcal{OS}(J_3) \rightarrow \mathcal{OS}(X)$ be a unital self-adjoint preserving map with $\varphi(J_3) = X$. Then the following are equivalent.*

- (i) φ is completely positive.
- (ii) φ is contractive.
- (iii) $\|\alpha X + \beta X^*\| \leq \sqrt{|\alpha|^2 + |\beta|^2}$ for all $\alpha, \beta \in \mathbb{C}$.

Farenick [11] proved that every positive map on the operator system of $n \times n$ Toeplitz matrices is completely positive. Therefore, Conjecture 6.10 equivalently states that every unital contractive map on the operator system generated by J_3 admits a positive extension to $\mathbb{M}_3$. We are not able to answer this at this stage. We leave this for further investigation in the future.

7. A unital contractive map on the operator system generated by a 4×4 normal

Figure 1: Conjecture 6.10 and its extension formulation in arXiv:2509.18027, p. 21.

The Counterexample

Theorem 1. *There exists $X \in \mathbb{M}_3$ such that*

$$\|\alpha X + \beta X^*\| \leq \sqrt{|\alpha|^2 + |\beta|^2} \quad (\alpha, \beta \in \mathbb{C}), \quad (1)$$

but the unital self-adjoint preserving map

$$\varphi : \mathcal{OS}(J_3) \longrightarrow \mathcal{OS}(X), \quad \varphi(J_3) = X,$$

is not 3-positive. Consequently φ is contractive but is not completely positive, and Conjecture 6.10 of [1] is false.

Lemma 1. *Let*

$$B_0 = \begin{bmatrix} 0 & 0 & 1/\sqrt{3} \\ \sqrt{2/3} & 0 & 0 \\ 0 & 1/\sqrt{3} & 0 \end{bmatrix}.$$

Then

$$B_0 B_0^* + B_0^* B_0 = \text{diag}(1, 1, 2/3),$$

and there is no Hilbert space $\mathcal{L} \supset \mathbb{C}^3$ and $C \in \mathcal{B}(\mathcal{L})$ such that

$$P_{\mathbb{C}^3} C|_{\mathbb{C}^3} = B_0, \quad CC^* + C^*C = I_{\mathcal{L}}.$$

Proof. The displayed identity follows by multiplication. Suppose such a C exists and write, relative to $\mathcal{L} = \mathbb{C}^3 \oplus \mathcal{L}_0$,

$$C = \begin{bmatrix} B_0 & A \\ D & E \end{bmatrix}.$$

The upper-left corner of $CC^* + C^*C = I$ gives

$$AA^* + D^*D = I_3 - (B_0B_0^* + B_0^*B_0) = \text{diag}(0, 0, 1/3). \quad (2)$$

Hence $A = e_3a$ and $D = de_3^*$ for some row operator a and column vector d over $\mathcal{L}_0$. The upper-right corner is

$$B_0D^* + AE^* + B_0^*A + D^*E = 0. \quad (3)$$

But

$$B_0e_3 = \frac{1}{\sqrt{3}}e_1, \quad B_0^*e_3 = \frac{1}{\sqrt{3}}e_2.$$

The e_1 -row of (3) is $d^*/\sqrt{3} = 0$, and its e_2 -row is $a/\sqrt{3} = 0$; the two remaining terms have range in $\mathbb{C}e_3$. Thus $a = d = 0$, contradicting (2). $\square$

Lemma 2. *Let*

$$\mathcal{G} = \left\{ G_{a,b} = \begin{bmatrix} 0 & a \\ b & 0 \end{bmatrix} : |a|^2 + |b|^2 = 1 \right\} \subset \mathbb{M}_2,$$

and let $\mathcal{K} = \text{mconv}(\mathcal{G})$, viewed as a matrix-convex set of pairs $(\text{Re } B, \text{Im } B)$. Then every level $\mathcal{K}(n)$ is compact, and $B_0 \notin \mathcal{K}(3)$.

Proof. At level n , the matrix-convex hull consists of

$$B = \sum_{j=1}^N V_j^* G_j V_j, \quad G_j \in \mathcal{G}, \quad V_j : \mathbb{C}^n \rightarrow \mathbb{C}^2, \quad \sum_{j=1}^N V_j^* V_j = I_n. \quad (4)$$

This set is compact. For completeness, normalize every nonzero V by $\text{tr}(V^*V) = 1$ and consider the compact set of pairs

$$(V^*GV, V^*V) \in \mathbb{M}_n \oplus (\mathbb{M}_n)_{\text{sa}}.$$

Equation (4) says that (B, I_n) belongs to the finite-dimensional conic hull of this compact set. Conic Carathéodory reduces every representation to a uniformly bounded number of terms, while the identity constraint fixes the total coefficient sum to $\text{tr } I_n = n$. The resulting parameter space is compact, so its image $\mathcal{K}(n)$ is compact.

If B_0 had a representation (4), set $C = \bigoplus_j G_j$ and $Vh = (V_1h, \dots, V_Nh)$. Then V is an isometry, $V^*CV = B_0$, and

$$CC^* + C^*C = \bigoplus_j (G_j G_j^* + G_j^* G_j) = I.$$

This contradicts Lemma 1. $\square$

Proof of Theorem 1. The matrix-convex set $\mathcal{K}$ is closed and contains zero: compressing any $G_{a,b}$ to its first coordinate gives zero at level one, and direct sums give zero at every level. Apply the Effros–Winkler matrix Hahn–Banach theorem [3, Theorem 2.2] at level three to the self-adjoint pair $(\text{Re } B_0, \text{Im } B_0) \notin \mathcal{K}(3)$. After changing signs of the coefficients if necessary, it gives $A_1, A_2 \in (\mathbb{M}_3)_{\text{sa}}$ such that the monic pencil

$$\mathcal{L}(B) = I + \text{Re}(B) \otimes A_1 + \text{Im}(B) \otimes A_2 \quad (5)$$

is positive semidefinite for every $B \in \mathcal{K}$, but $\mathcal{L}(B_0) \not\geq 0$.

Set $X = (A_1 - iA_2)/2 \in \mathbb{M}_3$. For every complex matrix B , direct expansion gives

$$\mathcal{L}(B) = I + B \otimes X + B^* \otimes X^*. \quad (6)$$

For $G_{a,b} \in \mathcal{G}$, positivity of (6) reads

$$\begin{bmatrix} I_3 & aX + \bar{b}X^* \\ bX + \bar{a}X^* & I_3 \end{bmatrix} \geq 0.$$

By the Schur complement, this is equivalent to

$$\|aX + \bar{b}X^*\| \leq 1 \quad (|a|^2 + |b|^2 = 1). \quad (7)$$

Given arbitrary $\alpha, \beta \in \mathbb{C}$, normalize $a = \alpha/r$, $b = \bar{\beta}/r$, where $r = \sqrt{|\alpha|^2 + |\beta|^2}$. Equation (7) gives (1); the zero case is immediate.

It remains to show that the resulting map is not completely positive. Lemma 5.2 of [1] states that

$$I_n \otimes I_3 + B \otimes J_3 + B^* \otimes J_3^* \geq 0 \iff BB^* + B^*B \leq I_n.$$

For $B = B_0$, the right side holds by Lemma 1. Thus the source pencil is positive in $\mathbb{M}_3(\mathcal{OS}(J_3))$. Applying $\text{id}_{\mathbb{M}_3} \otimes \varphi$ produces

$$I_3 \otimes I_3 + B_0 \otimes X + B_0^* \otimes X^* = \mathcal{L}(B_0),$$

up to the harmless tensor-factor flip. This is not positive by construction. Hence φ is not 3-positive and is not completely positive.

Finally, Theorem 5.1 of [1] proves that (1) is equivalent to contractivity of φ . Therefore φ is contractive but not completely positive. $\square$

Verification Notes

The script `code/check_finite_identities.py` verifies the defect of B_0 , equality for sample boundary generators, and the two saturated directions $B_0 e_3 = e_1/\sqrt{3}$ and $B_0^* e_3 = e_2/\sqrt{3}$. It is a sanity check only. The existence of X is supplied by the cited matrix Hahn–Banach theorem, not by numerical optimization.

The decisive external statement was checked in [3, Theorem 2.2]: if a closed matrix-convex set containing zero omits a point at level m , a monic linear pencil of size m separates that point from the entire set.

Scope and Novelty Check

This existential finite-dimensional counterexample disproves the implication (iii) $\Rightarrow$ (i), and hence the conjectured equivalence. It does not give numerical entries for X , nor characterize the full matricial range of J_3 .

The run indexes were searched for arXiv:2509.18027, the title, “ J_3 ”, “matricial range”, and “unital contractive map”. A bounded web search on 2026-07-22 used the exact phrases “Conjecture 6.10”, “Is every unital contractive map from $\mathcal{OS}(J_3)$ ”, the source title plus “counterexample”, and the arXiv id plus “Effros Winkler”. It found the source preprint and summaries but no later proof, disproof, or discussion of this argument.

Human Review Recommendation

Recommended verdict: likely valid counterexample. The two main review points are Lemma 2, which ensures separation applies, and the conversion (5)–(6). The dilation obstruction is proved independently rather than taken from the source’s brief “analogous argument” for arbitrary dilation size.

References

- [1] Pankaj Dey, Atanu Dhang, and Mithun Mukherjee, *Matricial ranges, dilations, and unital contractive maps*, arXiv:2509.18027, 2025.
- [2] Edward G. Effros and Søren Winkler, *Matrix convexity: operator analogues of the bipolar and Hahn–Banach theorems*, *Journal of Functional Analysis* **144** (1997), 117–152, doi:10.1006/jfan.1996.2958.
- [3] J. William Helton, Igor Klep, and Scott McCullough, *The tracial Hahn–Banach theorem, polar duals, matrix convex sets, and projections of free spectrahedra*, *Analysis & PDE* **11** (2018), 2343–2400; arXiv:1407.8198. Theorem 2.2 restates the Effros–Winkler separation theorem in the form used here.